

the corresponding private key. Also, it is impossible to derive the private key from the public key. Key lengths for asymmetric key cryptography are relatively longer (> 2048 bits). Examples: RSA, DSA, ECC, Diffie-Hellman algorithm.

Challenges with present cryptography

The existing cryptographic algorithms suffer from several shortcomings, such as:

- Intrusions in the system remain undetectable. There have been instances of huge financial and reputational loss when intrusions got unidentified for a long time.

- The security is based on the assumption that certain mathematical operations are impossible to be performed in considerable time, even by the most sophisticated classical computer. But security will be compromised with the advent of quantum computers.

- Brute force attacks on keys are also increasing due to the low frequency of key updates.

- Asymmetric encryption is 2-3 times slower than symmetric encryption due to longer key lengths and the complexity of the encryption algorithms used in asymmetric encryption. As one of the keys is public, both longer key lengths and complex algorithms are prerequisites.

Given the above facts, should we be worried? The answer is, yes. The report of IBM Security titled 'Cost of a Data Breach Report 2020' states the average total cost of a data breach is \$3.86 million, having an approximately 10% increase since 2014. In 2015, Michele Mosca presented a detailed overview of cybersecurity in the threat of possible advancements in quantum computing. He came up with an equation, popularly known as Mosca's inequality:

$$D + T \geq Q_c$$

where D stands for the amount

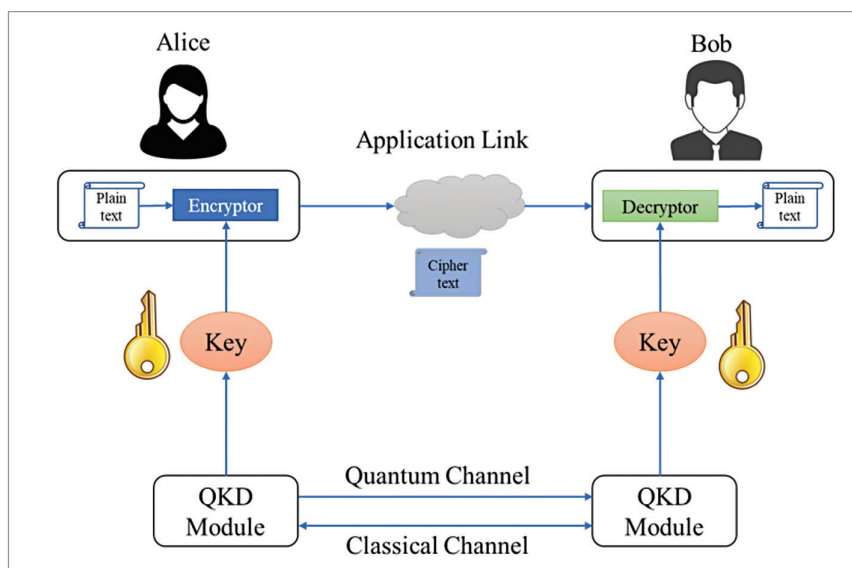


Fig. 2: P2P QKD link architecture

of time we need to keep our data secure, T the time it will take our systems to transition from classical to post-quantum, and Q_c the time it will take quantum processors to reach a scale where they can breach existing encryption protocols. There has been a consensus among experts that this inequality has already been breached in the year 2017.

QKD and its architecture

Quantum mechanics, formulated around 100 years back, is now maturing to an extent that its practical applications in the area of computing, communications, sensors, devices, etc have the potential to revolutionise these with unprecedented advances having far-reaching implications. QKD is one of the most emerging applications of quantum communications and will see a plethora of uses and deployments in the years to come.

QKD is a key sharing mechanism implemented in quantum cryptography to generate a perfectly random key that is shared between two nodes (traditionally called Alice and Bob). It is based on the principles of quantum mechanics, that is, Heisenberg's uncertainty principle and no-cloning principle, to guaran-

tee secure key distribution. These principles enable Alice and Bob to generate a shared random secure key that is known only to them. This secure key is used to encrypt the messages that need to be sent by Alice and to extract the original messages from received signals by Bob (and vice-versa). An important and unique feature of QKD is that both Alice and Bob can detect the presence of a third party trying to learn about the key, which inherently makes the generated keys highly secure from eavesdropping.

The sharing of keys between Alice and Bob takes place securely employing a quantum channel and an authenticated classical channel. The quantum channel for this system is typically either single-mode fibre or free space, whereas the classical channel can be a transmission over the internet. Quantum channel and classical channel together form a QKD link. The architecture of a point-to-point (P2P) QKD link is shown in Fig. 2.

The generation of shared secret keys through QKD involves several steps as follows:

- Alice (application sender) and Bob (application receiver) are connected through a classical channel